

Fast Forward

The Future(s) of the Cinematic Arts

Holly Willis



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Chapter 6

Virtual Reality and the Networked Self

In Alejandro González Iñárritu's gradiose film *The Revenant* (2015), a hunter named Hugh Glass (Leonardo DiCaprio) is left for dead by his fellow trappers in the wintery woods of Canada in 1823 after being mauled by a bear. The remainder of the film chronicles his travails as he struggles – nearly silent and often alone – initially just to survive in the vast expanse of snowy tundra, then to recover and, ultimately, to hunt down the man who killed his son. He tumbles over a cliff, flails through icy river rapids, and nearly freezes to death in a series of agonising physical trials, prompting critics to question the extremes of contemporary American cinema (see Lang 2015).

Both disturbingly violent and extraordinarily beautiful, the film serves as a compelling emblem for our current moment in the history of cinema in at once referencing the origins of cinema and pointing to its possible futures; in gesturing toward divergent media formats; and in auguring cinema's contributions to these forms, including virtual reality. Iñárritu has said, albeit with some contradiction, that he wanted to 'shoot this film in the most traditional and original of ways' and to honour that which affects him most as a filmmaker, namely, 'cinema itself: images over words. [...] This is the true power of cinema: visual storytelling' (in Hainley 2015: np). Indeed. If, during the dotcom boom, the mantra was 'content is king' in order to sustain the significance of some semblance of cohesiveness in storytelling craft in the face of the proliferation of distribution options introduced by the Internet, this emphasis on visual storytelling is the new incantation, repeated to distinguish cinema from virtual reality and a host of other emerging forms.

The Revenant was shot on a 65mm digital camera and the filmmakers employed an increased frame rate to better capture the available light in the film's Canadian locations, shooting for only a few hours each day. Reports about the film's production echo those of the plot itself, detailing the extreme conditions produced by very low temperatures, the constraints of shooting in far-flung locations and the hardships endured by

the film's cast and crew alike (DiCaprio reportedly opted to eat raw bison's liver for one pivotal scene in order to sustain the integrity of the film, for example) (see Miller 2015). These articles tend to underscore Iñárritu's desire for veracity, for the authentic. Indeed, in one article, Iñárritu states, 'We have lost the taste for the real' (Jonathan Romney 2015: no page). In short, then, Iñárritu's goal was to return cinema to the real, and in doing so, to rekindle our interest in it. This desire is embodied in everything from the triggering of a real avalanche toward the film's conclusion, to DiCaprio's breath which, on several occasions when the camera was close to the actor's face for a close-up, fogs the lens, blurring our view and insistently reminding us of the implacable physicality of both the humans in front of the camera, and the technology being used to represent them.

The result of these extraordinary measures is stunning. The snowy landscapes sparkle and glisten, the trees soar above us and the sky, with its expanse of white clouds, is breathtakingly vast. The camera swoops far above the landscapes, and then down among the trees with a thrilling and powerful grace. In this sense, the film is utterly cinematic, with 'cinematic' here referencing the awe-inducing powers of the moving image with regard to majestic cinematography and attention to detail; the expanse of the image in which sprawling landscapes were in many cases inhabited by a tiny human figure; and the fluidity of the camera, able to move with seamless grace within this tremendous space.

In a somewhat similar vein, in Ridley Scott's *The Martian* (2015), an astronaut is left for dead by his fellow astronauts and large sections of the film centre on his experiences, alone, trying to survive in the unfriendly environment of Mars. And similarly, in Alfonso Cuarón's *Gravity* (2013), we again find an astronaut alone, fighting for survival, in a film less interested in nuanced plot structure and traditional storytelling and dedicated instead to touting the extraordinary visual and technical capacities of cinema.

While *The Martian* weaves a conventional rescue narrative around the more experiential segments of the films – images of vast landscapes, the sky, and the body in its confined spaces – *Gravity* and *The Revenant* more assiduously render the lives of their protagonists alone, reducing narrative to armature, preferring to focus more specifically on embodied experience. Through Hugh Glass in *The Revenant*, we experience the bear's attack in visceral close-up and a wildly moving camera; we crawl with him on the snowy ground as he initially begins his pursuit; and we witness some of the most powerful scenes – the disembowelling of a horse – from a dispassionate distance. However, we also frequently just sit with Glass in the landscape. In each of these instances, the camera embodies not so much Glass's perspective but his emotional state, from the frantic terror of the mauling, to the almost manic determination to avenge a murder, to the pragmatic activities necessary to stave off death and finally to the more placid moments of wonder. As such, the camera serves as an extension of Glass's own perceptual system, inviting us to participate in his experience, and this, rather than the story, constitutes the logic of the film. It is as if we are not watching a

film tell us a story about something that happened in the past but instead experiencing events unfold with Glass, in the present.

In summary, with all three films, the awesome and extraordinary are aligned with the vastness of nature, whether in the wilds of the 1820s, or in more contemporary imagining of outer space. These films are stately celebrations of the cinematic *as* nature. In addition, these films relinquish the complexity of narrative structure and plotting, leaving them for the realm of televisual experience. In place of this narrative complexity, they turn to the depiction of affect and experience, and as such – in their attention to the seemingly boundless world all around us as witnessed by someone with greatly limited mobility – the films achieve an interesting thing: they sidle up to the technology gunning to overtake cinema, namely virtual reality. *Gravity* and *The Revenant*, engaging in a reduction of narrative storytelling and focusing on the plight of a single individual and his or her visceral experience: is this not the situation of the contemporary VR viewer who, with goggles on, stands alone, twisting and turning to view an immersive landscape that, in its expansive quality, can also produce a sense of wonder and awe?¹ (Or at least, that's what it is intended to do – so many current VR projects still fail in this effort!)

But why would these films emulate VR? In part, they may simply strive to suggest that cinema can still render an extraordinary visual experience better than the clunky visuals currently available for view through uncomfortable goggles in the early stages of twenty-first century VR. In some cheerfully contradictory way, they achieve the experience of the virtual through the cinematic.

However, the films also function as fantasies. In *The Cinema Dreams its Rivals: Media Fantasy Films From Radio to the Internet* (2006), Paul Young explains his concept of media fantasy films, that they 'are expressions, in different periods and under different conditions – conditions determined by differences in new rival media and the fantasies they sponsor – of an ongoing institutional defence of classical form, address and spectatorship' (2006: xxviii). While Young argues that every film in some ways makes a claim about its medium, media fantasy films most explicitly grapple with the sense of rivalry as media forms jostle for power at critical historical moments. To continue the enquiry into the role of *Gravity* and *The Revenant* in relation to virtual reality, both serve as media fantasy films in blurring the boundaries between classical Hollywood filmmaking and VR, offering the intensive experience of and identification with a singular protagonist characteristic of cinema, while at the same time allowing us to inhabit – and almost explore – a vast landscape, which is one of the primary tropes of contemporary VR. Taken together, the two films also span the polarity of cinema, from animation to live-action. Where *Gravity* touts its virtual constructedness and has even been dubbed an animation due to its extensive use of CG, *The Revenant* insists on its cinematic heritage. In this way, 'cinema' covers an incredible territory.

In this chapter, I will explore several re-emerging media forms, including 360-degree cinematography, stereoscopic cinema and virtual reality, but to get there, I will start with the role of the mobile and increasingly virtual camera in cinema, which I believe

serves as a precursor to VR. I will also address the topic of presence, so much an arena of interest in contemporary life. And I will bring these ideas together to query the role of narrative in these forms, as we experience a cultural moment in which narrative, if not waning, is surely morphing into new forms or being supplemented and challenged by other storytelling structures. As such, the work I will address responds to the shifting needs of a culture characterised as both posthuman and post-cinematic. What place do character, identity, temporal and spatial coherence have in these works?

The Camera Continuum

Perhaps one of the most striking characteristics of digital cinematography is its new-found mobility and capacity for shot duration. Camera mobility was greatly enhanced with the introduction of the Steadicam in 1975, allowing filmmakers to communicate spatial transitions in single takes by moving the camera fluidly through space. Whereas spatial organisation in earlier filmmaking was established through cuts or deep focus, or on occasion, via split-screen techniques, the Steadicam enabled the ability to communicate spatial positioning via movement (see Schraa 2007). Many contemporary films expand on the fluidity of the Steadicam, jettisoning the sense of the camera as the bulky, heavy object it once was; we witness a weightless manoeuvrability as the camera effortlessly traverses spaces. However, often, what we are witnessing may not be the movement of a physical camera but instead the point of view captured by a virtual camera moving through a body of information. The experience of mobility was exponentially increased with computer-generated imagery (CGI), which not only made movement easier, but allowed the 'camera' to travel places that in the past were physically impossible. As Michael Schraa points out,

A virtual camera can extend into spaces that are impossible or prohibitive for a physical camera. Virtual cameras can go through walls; they are not hindered by gravity or tied to a means of physical propulsion. They also offer unprecedented control over camera movement so that all 360 degrees of rotation are made available at any given speed. (2007: np)

Virtual cameras are not especially new – they were presented at SIGGRAPH in the late 1990s – but they have become far more prevalent in contemporary Hollywood filmmaking, in part due to their familiarity within the context of game design and play.²

Virtual cameras often make use of motion or performance capture. For several years, USC's School of Cinematic Arts has been lucky enough to have a 46-camera Vicon System to use for performance capture, and it was here that I first began to understand the shift in filmmaking paradigms. The system consists of an array of the cameras placed on a grid around the perimeter of a large, high-ceilinged room. In the middle of the room is a grey square marked on the carpeted floor, which designates the 'volume' that can be captured by the camera array. Performers clothed in black

suits with reflective markers perform inside the volume, and their actions are captured from 46 different perspectives. This information is fed into a computer, where it can be used to have the performer appear to be something else – an animated dinosaur, for example, with the markers capturing specific reference points in space. In this case, the performer ‘drives’ or performs another character, which is seen onscreen; or the performance within the volume can be stored and later placed in a sequence. In either case, at a later point, the material can be ‘navigated’ by a virtual camera. The camera moves can be explored, practiced and repeated until the right ‘shot’ has been crafted.

In explaining the system, Eric Furie, the School’s Digital Systems Specialist and the instructor for a performance capture class, explains that in traditional live-action filmmaking, production integrates the directing of performance, lighting, make-up, camera work and all of the other crafts simultaneously. However, with motion capture, these components are broken down and treated individually. In addition, directing becomes more akin to theatrical directing. ‘The volume is essentially a black box theatre,’ Furie observes. ‘The lighting can be changed later, as can the make-up. If I want to include the camera perspective at the same time as a reference – and there are benefits to doing that as most actors like to know where the camera is – I can do that. But there is nothing to prevent me from coming in after I’ve recorded the performance and “shooting” the scene thousands of times over until I get the shot that I want’ (in Willis 2011: np). Furie goes on to explain that motion capture essentially merges production practices, animation techniques and interactive design. ‘It steals from each, and feeds back into each’, he says (*ibid.*).

The Vicon System is akin in some ways to the Light Stage systems that emerged from research by Paul Debevec at UC Berkeley and later at USC’s Institute for Creative Technologies beginning in the late 1990s. Through an iterative process, the Light Stage system has evolved from early attempts to light a human face from all directions in order to acquire what Debevec calls the ‘4D reflectance field of the actor’ (2012: np.) to stages that integrate a lighting array that captures multiple facial expressions in diverse lighting situations. The third Light Stage used 156 coloured lights distributed across a geodesic sphere that allowed the imagery captured through the system to be composited with already existing scenes; the innovation here was that the Light Stage imagery would match the original imagery’s illumination. The sixth Light Stage is large enough to illuminate an entire human body. It employs 6,666 led lights and features a rotating treadmill. A project titled ‘Relighting Human Motion’ (2006) was designed to capture not just the human face but the body in motion. According to Debevec, it ‘recorded actors walking and running on a rotating treadmill inside the stage at 990 frames per second, under a basis of 33 repeating lighting conditions from a vertical array of three high-speed cameras’ (*ibid.*). The goal for this iteration of the Light Stage was to create imagery that could be integrated seamlessly with existing footage. The material produced by the various Light Stages has been used in a long list of films, including *Spider-Man 2* (2004), *King Kong* (2005), *Superman Returns* (2006) and *Avatar*, among others.

What is significant about the Vicon and Light Stage systems is the degree to which they posit a different logic of image capture from that of the traditional camera with its associated filmmaking practice. Rather than a single camera, or even two or three cameras, capturing a scene from specific perspectives, both systems enable the capture of an entire space and the objects within it. The 'camera' move can then be created from that information. As David H. Fleming and William Brown point out in reference to James Cameron and *Avatar*, 'We have here, then, a complete reversal of traditional filmmaking: rather than build a set or find a location through which the camera then moves, Cameron instead moves his camera through low-definition images that are then made into high-definition images for the finished film' (2015: para 25). Here we have a shift away from the vision of the filmmaker, and by extension the human, to an all-encompassing vision that flattens the hierarchies established so assiduously with the traditional camera.

Returning to traditional filmmaking practices, the mobility exploited by the virtual camera was presaged by the growing manoeuvrability of the physical camera and a visual style characterised by motion, whether fluid and achieved through longer takes, or chaotic and achieved through rapid cutting. David Bordwell has identified what he dubs an 'intensified continuity' in response to the sense that many contemporary films appear to lose the temporal and spatial coherence created through cinematography and editing characteristic of the classical Hollywood style. He argues, however, that these films represent not a break with past visual forms of temporal and spatial structuring within narrative but a continuation:

Far from rejecting traditional continuity in the name of fragmentation and incoherence, the new style amounts to an *intensification* of established techniques. Intensified continuity is traditional continuity amped up, raised to a higher pitch of emphasis. It is the dominant style of American mass-audience films today. (2002: 16)

Bordwell goes on to identify four specific strategies through which this intensification occurs: rapid editing, with shot lengths growing steadily shorter; bipolar extremes of lens lengths; a reliance on close shots; and wide-ranging camera movements. He further develops these four strategies in *The Way Hollywood Tells It: Story and Style in Modern Movies*, where he also describes the 'prowling camera' and the prolonged following shot 'in which the camera tracks a character moving along a lengthy path' (2006: 134). Bordwell traces this use of the camera from the 1930s, with films such as *Scarface* (1932), through the 1950s and *Touch of Evil* (1957), up to contemporary music videos, such as that for Björk's 'Hidden Place' (2001) and he highlights the fact that cameras during the first decade of the twenty-first century move more than ever, and suggests that this is due in part to the desire to boost a scene's energy: 'Camera movement also separates the planes of the image and creates a more voluminous pictorial space' (2006: 136). Finally, he cites cinematographer Vilmos Zsigmond, who explains

that this technique ‘gives you the third dimension, which is the way movies should be. If you lock down the camera, it’s like looking with one eye’ (2006: 137).

While Bordwell insists that the roving camera, as well as the long take, continue stylistic choices established decades ago, the use of the long take in a film such as Iñárritu’s *Birdman* (2014), with the majority of the film captured in what appears to be one, single, fast-paced shot, exemplifies the desire to bring the audience into the protagonist’s temporal and spatial experience both intimately and haptically. It also represents an aesthetic in which we vividly sense that a world exists before us, and the camera is tracing a path through that space, a space that is seemingly continuous. Rather than taking the continuous space of the profilmic event and breaking it into shots which are reassembled to guide viewers to experience a scene in a particular way, the prowling camera offers a pathway through a continuous space, with point of view attributed either to a character or to something else, perhaps something in the world. As those who have dedicated their filmmaking practice to world building assert, the world of the story gains in significance in this mode of filmmaking.

Akin to *Birdman*, Gus van Sant’s *Elephant* (2003) similarly includes very long tracking shots; in this film, although the camera moves far more slowly than that of *Birdman*, it traverses a story space in a manner reminiscent of first-person shooter video games, following just behind specific characters and visualising the world very close to what the avatar in the game might see. In this way, the world in the film seemingly exists for us to explore. Gaspar Noé’s *Enter the Void* (2009) takes this tendency to an extreme, with a camera that initially assumes a first-person perspective – we even see the character’s eyes blinking – to become an entirely disembodied camera capable of impossible perspectives, at points floating high above the city and moving with breath-taking fluidity.

This shift from one visual logic to another is charted by William Brown in *Super-cinema: Film-Philosophy for the Digital Age*:

Although Edward Branigan ... and Stephen Prince ... argue that image conventions in digital cinema remain, broadly speaking, perspectival (in that they mimic human perception), the changing scales of digital cinema, which shifts from the micro (brain cells, bacteria in drops of water) to the macro (galaxies), and which conflates inside and outside, in fact suggest that digital cinema moves beyond human perspective – and not just across shots (something cinema has long since been able to achieve), but in single shots that combine these perspectives as if they were a part of a single continuum. (2013: 1141)

Brown highlights the camera movement in the opening moments of *Fight Club* (1999), where the film ‘shows a continuity of space that is digitally enabled and posthuman, in that the camera passes through empty space as well as the objects that fill it with equal ease’ (2013: 928). He continues: ‘And in *Fight Club*, the posthuman continuity that sees humans, objects and empty space all seemingly sharing an equal ontology, in that

the (virtual) camera does not distinguish between them, offers us, as a result, a conception of space that is different from the fragmented spaces of analogue cinema' (2013: 936). Notably, *Fight Club* is touted as the first film to use photogrammetry, mapping together physical and virtual spaces to create a composite.

The difference that Brown begins to sketch points to the imbrication of cinematic and game spaces which come together in virtual reality, along with a host of technologies that map representational and informational spaces together, with the 'spectator' becoming at once an agent within a continuous space who sees, but more significantly, is seen and tracked. The films and videos that deploy the long take and the prowling camera prepare us for an emerging paradigm within which we merge with the world around us in a posthuman decentring deftly aligned with a culture dedicated to producing subjects as interfaces, subjects as elements within a network. The speculative implications of this merging are vast. They suggest a compelling model of posthuman storytelling experience that does not presume the human as the central axis for both agency and perspective. They illuminate possibilities for imagining the world as an ecosystem within which the relations among all things constitute a primary force, and a potential grounding for story through affect and senses beyond the visual. However, they also point to the ways in which the network becomes the centre, and being tracked, we serve the system. As Alexander Galloway and Eugene Thacker write of individuals within a control society,

It is a question not of being individuated as a 'subject' but instead of being individuated as a node integrated into one or more networks. Thus one speaks not of a subject interpellated by this or that social force. One speaks instead of 'friends of friends', of the financial and health networks created by the subject simply in its being alive. (2007: 60)

The task, then, is to imagine a productive form of individuation as a node within a network, one that is not reducible to our, often inane, interactions online.

What Is a Camera?

And what do we call the camera if it is no longer actually a camera? Ryan Pierson tackles this question in 'Whole-Screen Metamorphosis and the Imagined Camera (Notes on Perspectival Movement in Animation)' (2015). He begins by describing the movement of the camera in a sequence that occurs early in *Gravity*, when Ryan Stone (Sandra Bullock) is floating in space.

As Stone is drifting, she is tumbling over herself like a rotating planet, and as the camera approaches her, it begins, seamlessly, to tumble with her; approaching closer, it passes through the glass of her helmet, studying her eyes as they dart frantically; approaching even closer, it turns itself around to look outward, darting

about as Stone's eyes were just doing, at the space that it was comfortably occupying a moment ago. (2015: 7)

The camera, then, both achieves a kind of mobility that would be impossible for a physical camera, and it provides a perspective that at one moment is decidedly disconnected from a particular viewing subject and then comes to inhabit a first-person point-of-view. Pierson goes on the query the conundrum of what he dubs 'the camera in quotes', writing that the camera has become more 'animated':

I mean 'animated' here in two senses: in the technical sense that it is produced synthetically rather than recorded photographically, and in the sense that it has the kinds of strange powers we normally attribute to cartoon characters. Consequently, we don't quite know how to talk about the moving camera. (2015: 7)³

Pierson here identifies one of the fundamental qualities associated with contemporary filmmaking practices in which shifts in digital tools remain so far removed from the working methods of many filmmakers that we lack the language to even describe or define new practices. As live-action and animation blur together, and as previously dissimilar practices – such as those of film production and game design – come to resemble each other, the language must evolve as well.

Mike Jones also addresses this dilemma in 'Vanishing Point: Spatial Composition and the Virtual Camera' (2007), once again within the context of animation. 'The entity of the camera becomes not an apparatus of moving image capture but rather a spatially specific vanishing point of perception' (2007: 228). Further, 'The physical camera has long been associated with the "eye", as a simulation of the "eye" in an alternate world (either anthropomorphically or otherwise) and in doing so is rooted to a physical and tangible exploration and depiction of space that obeys rules of physics' (ibid.). He adds in contrast that the virtual camera 'moves beyond this into new conceptualizations of space and the viewer's connection to, or immersion in, it' (ibid.), and suggests that the virtual camera 'becomes a simulation of "I" rather than "eye"; a simulation of viewer-derived presence in space rather than an anthropomorphically based viewing apparatus' (ibid.).

Jones explores several examples of the seemingly omnipotent power of the virtual camera, citing the opening shot of *Angels in America* (2004), as well as the impressive shot in the opening scenes in *Fight Club*, and a shot in *Panic Room* (2002) in which the camera travels in one apparently seamless take through a stairwell, into a room, across a table and ultimately inside the lock on the door. As Jones points out, these cameras ignore the rules of diegetic space. 'The camera does not share the spatial conventions or restrictions that all other depicted cinematic elements such as characters, environs, physical objects and events in the scene do' (2007: 236). He goes on to make an intriguing point using the example from *Panic Room*: the camera's perspective, in moving beyond the rules governing physical space, must inhabit a different spatial

logic. The camera, he writes, 'can be viewed as a point-of-view shot from the perspective of the house and its holistic internal space' (2007: 237). The so-called camera does not *occupy* space; it is *of* the space: 'The POV of the virtual camera's exploration itself becomes an act of "seeing what the space sees" rather than what the characters can or might see' (ibid.).

This assessment of the role of the camera clearly has very significant implications for the ways in which viewers watch films, shifting away from forms of cinematic identification produced in previous generations of cinema viewership in which shot and editing conventions created a complex but stable system:

This can be seen as a substantial and even dramatic shift in cinematic awareness of experience whereby we move from a position of cinema as a scenic/spatial simulation of the abstracted 'eye' to a fundamentally new conceptualization that is, in essence, an emulation of a spatially abstracted 'I' – a consciousness of presence, a spatially removed perspective but one nonetheless intimate and evoked as non-diegetic viewer observation. (2007: 238)

The camera, in this logic, no longer represents an authorial perspective, he argues. It does not communicate a story through the logic of 'this is what happened'. Instead, it communicates in the present, 'this is happening'. Rather than being shown something, we simply see. Jones next links this experience to immersion. However, we might extend the notion of 'seeing what the space sees' to the posthuman, which requires a similar decentring of the human, and we might, again, link this emphasis on the environment with the logic of virtual reality, which presumes a viewer primed to discover his or her own point of view.

The challenge for so many of us interested in image-based storytelling is fathoming a shift from one system of representation to another. Thomas Elsaesser has written lucidly on digital cinema and the virtual, and outlines precisely this conceptual challenge:

The difficulty (and confusion) arises because in the case of (photographic, analogue) cinema, we are operating within an ocular-perceptual frame of reference (where technical images only know an ontology of realism/illusionism, real/fake, and whose epistemology is true/false). Within this field, we are bound to see the virtual (as in VR) as part of the ocular-perceptual register, which corresponds to the logic of photographic cinema, regardless of whether we see its representational mode as 'perspectival', 'flat', or 'multi-focal'. Thus, in the digital, we do not have an ocular-perceptual ontology; instead we have a digital ontology, which is that of the dynamically variable, scalable. (2014: 300)

He goes on to suggest that the logic of the virtual may end up simply absorbing the logic of the cinematic. He writes provocatively that as this digital logic extends cultur-

ally to become pervasive, we will experience a shift in temporality: 'The virtual in this sense may even create a modality, a temporality, a tense, which the visual does not know at all, but which brings the virtual closer to language, rhetoric and formal logic than one might initially supposed' (2014: 301). He continues:

In addition to the (algorithmic) logic of 'if ... then' conditionals and the hypothetical mode 'what ... if', one would also want to argue that one of the implications of this new cultural logic of the virtual is the prevalence of the tense of the future perfect. The future perfect is a tense that refers to a completed action in the future. When one uses this tense one is projecting oneself forward into the future and looking back at an action that will be completed some time later than now. (Ibid.)

Elsaesser goes on to argue that VR operates through a fantasy of the real, despite the fact that it is based on the body's movements (or at least the movement of the head); as such, it requires a grammar, which means that it 'belongs to the realm of the semiotic as much as the phenomenal' (2014: 302). This leads him to state that 'the virtual is a tense, namely, that of the conditional' (ibid.). What does it mean for the virtual to be in the tense of the future perfect? Here Elsaesser captures the underlying logic of the algorithm; however, he also illuminates the ways in which the experience produced within a VR environment is generative and dependent on the actions of the participant.

Before moving on, I want to highlight one last point about the disappearing camera. David H. Fleming and William Brown use the notion of the skeuomorph in 'A Skeuomorphic Cinema: Film Form, Content and Criticism in the "Post-Analogue" Era' (2015). For Fleming and Brown, the skeuomorph refers to 'an object or form that anachronistically retains ornamental features or design cues from an earlier technological era or method of production, and which no longer have any functional purpose' (2015: para 4). Katherine Hayles (1999) has used the example of fake stitching on a vinyl car seat that visually resembles the real stitching in leather seats but, in the vinyl, is functionally useless. It serves only as a comforting reference to a past object. In contemporary cinematography and the shift from analogue to digital, rather than celebrating the differences between the two modes, the goal more often has been instead to emulate the analogue by ameliorating evidence of the digital. The early history of digital cinema was based precisely on this desire to overcome what were deemed the inadequacies of the visual quality of video.

However, the skeuomorph functions in a more specific way relative to filmmaking equipment as the devices used to capture visual information begin to take new forms and shapes. These devices are not cameras in the same sense that film cameras were cameras. They may not be large box-shaped devices with long, round lenses. They do not record, but instead capture. And there is no substrate onto which anything is recorded. One of the devices used on *Avatar*, for example, was called a 'swing-cam' and while it was an entirely digital recorder, it was explicitly devised to resemble a twentieth-century film camera. The device, however, was radically different in form

and function from the object it was designed to emulate. It had no lens, for example, and rather than recording images or scenes, it was instead able to ‘navigate’ through the three-dimensional volume within which it was capturing information.⁴

Adapting to that paradigm is challenging for many artists, however. Fleming and Brown use the notion of the skeuomorph to suggest the implications of this challenge: ‘Indeed, many of the skeuomorphic features that contemporary digital cinema retains/displays highlight the aetiological or atavistic link between digital cinema and its twentieth century, analogue predecessor; these links being for reasons of “fashion” rather than “function”’ (2015: para 9). This in turn leads to an impoverishing of abilities in understanding the digital. ‘Skeuomorphic features of digital cinema are also deceptive in that they disguise the true nature and power of this precisely new medium as they diverge and diversify’, note Fleming and Brown (ibid.), who go on to assert that contemporary audiences and filmmakers have not gleaned the ability to understand emerging forms.

The question before us now, then, is to what extent contemporary VR experiences are themselves skeuomorphic? Just as *The Revenant* and *Gravity* overlap with VR experiences, how are the tropes and features of the cinematic inflecting and constraining VR? As we move from leather to vinyl, what can be discovered in the new material?

Virtual Reality

‘Virtual reality did not spring, like Athena from the forehead of Zeus, full-blown from the mind of William Gibson,’ Katherine Hayles wrote (1996: 11), gently reprimanding those who might too easily designate a single impetus or source for something as manifold as virtual reality. Her comment references Gibson’s ‘cyberpunk’ novels from the 1980s, including *Neuromancer* (1984), which helped imagine the near future through fiction, and introduced many of the key concepts, including cyberspace, that augured our culture’s introduction to the Internet.

Today, we might feel obliged to dutifully echo Hayles’s comment and say that the next generation of virtual reality did not spring from the mind of Palmer Luckey, who, in 2012 at the age of 19, sold his hand-made VR headset to a group of investors, who in turn sold it to Facebook in March of 2014 for \$2 billion, launching a new round of enthusiasm for VR and the development of tools and experiences in the medium. The reminder Hayles puts forward unites these stories, however, underscoring what should be obvious, that the evolution of a technology is always far more complex than simply the desire or imagination of an individual. Further, the ‘new’ is often understood as such only through wilful amnesia in which the publications, technologies, projects and critiques from the past rest undisturbed. This is especially true in 2016, when public relations efforts to celebrate the arrival of VR require an amazing degree of repression and the maintenance of a particular idea of the viewing subject from another era, all of which we might generously attribute to the exigencies of a world known for its precarity; a future characterised as unknown but most certainly unfamiliar; and

a sense of self quickly losing its uniqueness and sovereignty as the things around us gain cognition. From this perspective, the stories of Luckey's luck that have graced innumerable magazines in 2014 and 2015, including *The Smithsonian*, *Time*, *Vanity Fair* (with photography by Annie Leibovitz) and *Forbes*, among others, are predictably giddy and understandably repetitive tales that, disregarding Hayles's plea, imagine the origins of VR springing fully formed from the mind of the barefoot teenager. (Indeed, *Time*'s cover image from its 6 August 2015 issue featuring Luckey leaping up from the sands of a California beach while wearing his Rift goggles was so silly that it was widely lampooned and humorously reimaged.) However, what this playful narrative reinscribes in the context of VR is the significance of the individual, with Luckey's Oculus Rift offering a tool that, at least to begin, imagines singular viewers wearing singular headsets in a logic that remains centred on the eye at its centre, on the ocular regime. The point I want to make here is simple: the story of VR that currently resides at the forefront of cultural consciousness is of a boy genius who became a billionaire with a tool devoted to the eye. This story not only forgets the near past, but obscures the radical implications of VR, which centre precisely on a very different story, namely that we have entered a new paradigm characterised not by the singular individual and the photographic, but by the collective construed in relation to context, the network and a frame of reference made possible by the posthuman.

Definitions

Virtual reality has been defined and described in many ways, as have a host of terms coincident with VR, including embodiment, presence and immersion. The histories of VR invariably begin with the nineteenth-century panoramas comprised of 360-degree painted scenes (see Grau 2003). In 1999, in a major assessment of VR projects and initiatives nationally, Frederick P. Brooks defined VR as an experience in 'which the user is effectively immersed in a responsive virtual world. [...] This implies user dynamic control of viewpoint' (1999: 16). In 2016, the term has become quite diffuse, especially as companies and their systems – the Oculus Rift, Sony's PlayStation VR and the HTC Vive, for example – compete for domination in the entertainment market, while organisations as diverse as *The New York Times*, with its NYT VR initiative; VRLA with its bombastic tradeshow events; and the New Museum's 2016 Triennial titled 'Surround Audience' each suggest slightly differing definitions of and visions for the term.

Generally, however, a VR experience includes the following elements:

- a display system that delivers computer-generated directional sounds and images to a participant. Brooks adds that the visual display should 'block out contradictory sensory impressions from the real world' (ibid.);
- a graphics system that can generate images at a high frame rate, generally 90 fps;

- a tracking system so that the participant's head movement and directionality can be tracked, along with hands and body in some systems;
- and often some kind of haptic feedback system, which might include gloves or handheld objects which can be tracked and visualised on screen.

VR systems at the moment include those experienced through head-mounted displays, the cumbersome device placed over the head with a mask akin to what one wears for undersea diving; and CAVEs, which stands for Cave Automatic Virtual Environment. CAVEs were invented by researchers in the Electronic Visualization Lab at the University of Illinois in 1992. They are rooms in which rear-screen projections on the walls create an experience of immersion. The cave experience can accommodate more than one person and also requires that participants wear stereo glasses that control the images for each eye. At present, because of the emphasis on the commercial marketability of VR for in-home entertainment, CAVE systems as well as other elaborate, large-scale, rear projection apparatuses such as Tsukuba University's Large Space, occupy a different realm of the VR industry that is largely confined to universities and research facilities.

In a quick, now canonised list of VR's inventors, Ivan Sutherland is credited with creating the first VR prototype. He also wrote a key essay in 1965 entitled 'The Ultimate Display' exploring the design for a kinaesthetic display that would attend to multiple senses. He writes, 'If the task of the display is to serve as a looking-glass into the mathematical wonderland constructed in computer memory, it should serve as many senses as possible' (1965: 1). Myron Krueger is credited with first using the phrase 'artificial reality' in a 1974 dissertation which was later published as a book in 1983, while Jaron Lanier, of VPL Research (VPL stands for Visual Programming Language) is credited with using the term 'virtual reality' first. Morton Heilig developed the Sensorama system in the 1950s, which is widely regarded as a precursor to VR in its aspirations. The work done at NASA Ames was also significant. Scott Fisher, who had worked with Alan Kay at Atari as a research scientist, joined the Ames Research Centre in 1985, and until 1990 served as the founder and director of the Virtual Environment Workstation Project (VIEW), which is credited with innovating many of the first VR technologies, including 3D audio technology, head-mounted displays with stereoscopic imagery and an adaptation of the dataglove. His essay, 'Virtual Environments, Personal Simulation and Telepresence' (1991) describes a world in which people will easily connect, communicate and collaborate through what Fisher dubbed 'telepresence' and VR systems. Fisher's vision for VIEW, and VR generally, was similar to that of his predecessors in wanting to include multiple senses. In this regard, Fisher was inspired by Heilig, a cinematographer whose vision of an immersive movie experience was one of the driving visions for VR. The Sensorama was patented in 1962 and resembled an arcade game that offered participants not simply a short movie experience, but one in which the seat moved, odours surfaced and the wind blew as was appropriate to the images on display. Heilig was never able to sell

his device, so it languished, but his vision of a multi-sensory movie experience lives on. Heilig also developed an early head-mounted display called the Telesphere which resembles many of the current HMDs (see Jordan and Packer 2002; Delaney 2014).

Some of the written texts from the 1990s on VR also remain surprisingly timely. Howard Rheingold's *Virtual Reality: The Revolutionary Technology*, for example, describes VR in this way:

Imagine immersing yourself in an artificial world and actively exploring it, rather than peering in at it from a fixed perspective through a flat screen in a movie theatre, on a television set or on a computer display. Imagine that you are the creator as well as the consumer of your artificial experience, with the power to use a gesture or word to remould the world you see and hear and feel. (1991: 16)

In addition to articulating the key element of VR – namely that ‘you are the creator as well as the consumer’ – Rheingold also identified the three qualities that characterise VR in a list that continues to inform most contemporary descriptions, including the one above. His three qualities include immersion; the ability to walk around in the three-dimensional space of the project while choosing your own point of view; and the ability to manipulate the world in some way.⁵

VR in the 1990s was a rich, robust area of exploration – in technologies, story structures and experience design. To be sure, it was also an era rife with hyperbole about both the potentials and threats of VR. As Ellen Strain argues in ‘Virtual VR’ (1999), the elegant cinematic depictions of VR oversold the technology to audiences who were then dismayed when they encountered the clunky reality of VR in the 1990s. Steve Anderson echoes this thesis in his online archive ‘Technologies of Cinema’, which aggregates dozens of clips from film and television that illustrate how VR was depicted in Hollywood.

Several of the most significant projects of the 1990s were designed by women. *Placeholder* (1992), by Brenda Laurel in collaboration with Rachel Strickland, Rob Tow and Michael Naimark, for example, is widely lauded for its thoughtful and productive exploration of the potentials of VR for crafting a narrative experience. The project networked two participants, who were then directed through audio by a persona designated as ‘the goddess’ during the experience. The piece included three different environments – a cave, a waterfall and a river – based on three locations near Banff National Park in Alberta, Canada. The participants could become any one of four animal characters – spider, snake, fish and crow – chosen from aboriginal tales, and they could wander within the space of the project and explore. However, it was not until they had intersected with one of the animals that they would begin to enjoy a richer experience; once the intersection occurred, participants experienced a shift in movement, vision and perception.

Placeholder served as a powerful prototype for exploring presence, the role of the body, and world building, that is, the design of a narrative within a virtual space. It also

proceeded by first creating the environment of the story, understanding that in VR, stories literally ‘take place’. In their expansive description of the project and its goals, Brenda Laurel and Rachel Strickland note:

We thought of *Placeholder* as a set of environments imbued with narrative potential – places that could be experienced and marked through narrative activity. When a person visits a place, the stories that are told about it – by companions, by rock art or graffiti, or even by oneself through memories or fantasies – become part of the character of the place. Stories give us ideas about what can be done or imagined in a place; learning that a particular canyon was an outlaw’s hiding place, for instance, or remembering a child saying that a particular rock resembled an old woman’s face will certainly influence our experience of that place. It’s hard to experience a natural place without remembering or constructing some stories about it. (1994: 124)

In this description, the role of the storyteller working in the context of VR is not necessarily to tell a story but to craft a rich and robust world within which stories can occur; these are stories that are then made, rather than told, by participants through their experience in the world. The onus of storytelling thus shifts from an author, writer and director to the participant; the work of the artist is to craft a world worthy of participation.

Placeholder is but one example of the powerful experiments with VR in the 1990s, which also include the work of Char Davies with *Osmose* (1994–95), which used the breath as a means for navigation, and *Éphémère* (1998), as well as Yacov Sharir and Diane Gromala’s *Dancing With the Virtual Dervish: Virtual Bodies* (1994). Each of these projects relied on blurring the boundaries between abstract and concrete, presenting landscapes and movements that were at once ethereally detached and deeply embodied. Rather than striving for live-action experiences as extensions of the cinematic, these projects began with the body and queried its role in the production of meaning.

VR was also theorised in the 1990s, with hearty critiques alongside technology-oriented research papers. Simon Penny, for example, wrote an essay titled ‘Virtual Reality as the End of the Enlightenment Project’ following what he describes as ‘the now legendary Virtual Reality panel at Siggraph ’91’ exploring the cultural context that gives rise to VR and pinpointing a collection of key themes that continue to resonate today, namely embodiment within the context of virtual reality, the experience of space, cognitive plasticity and the politics of interactivity. Simon Penny writes, ‘The developers of VR have (unwittingly?) inherited a humanistic world view (an attitude to life and a way of making pictures) which places the eye of the viewer in a position of command, a privileged viewpoint on the world’ (1992: np). He goes on to ask a series of questions that remain: ‘What if VR had developed along pictorial principles other than renaissance humanism?’ and ‘What if VR developed in a culture with a different attitude to the body?’, concluding with an arch description of VR in the 1990s:

This readiness for VR has been prepared (most recently) by Disneyland, Hollywood, Liposuction and Nintendo. Conceptually vacuous theme parks, anesthetizing cinema, interactive games that perpetuate the myth of the individual and the cult of violence-as-liberty. And perhaps most significantly, the acceptance that the body may be customized at will like some kind of hotrod. (Ibid.)

Other scholars voice concerns as well: Sandy Stone queried the role of the body in 'Will the Real Body Please Stand Up' (1991), for example, and Stephen Wilson captures many of the major questions from that era in chapters six and seven of *Information Arts: Intersections of Art, Science and Technology* (2002). However, what should be clear already is that some of the first attempts to construe virtual experiences were motivated by a deep interest in the body and its permutations; by a reimagining of the parameters and even definition of space; and by a posthuman desire to consider the world from a non-human perspective.

Mapping Immersive Video

Like other genres of the post-cinematic explored so far, contemporary virtual reality hovers at the intersection of multiple realms, with applications in the military, medicine and science, as well as cinematic storytelling and gaming. Indeed, game designers are perhaps the most comfortable with design for VR due to their familiarity both with 3D design and navigation, as well as with procedurally generated environments. However, for filmmakers, VR represents a radical shift in orientation, whether they opt to work in animated space or live-action; in either case, the structuring of time, space, perspective, identification and narrative all require reimagining.

Steve Anderson (2016) has helpfully mapped a technology-agnostic array of immersive platforms and projects clustered under the VR rubric in terms of degrees of interactivity, fixed or elastic time bases, the ability to procedurally affect project content, and distinctions among lens-based and computer-generated imagery.⁶ While many still strive to create the full-body sensory immersive experience imagined by Heilig, others are exploring more agile or casual experiences that deploy common devices such as tablets and mobile phones. In order to speak coherently about the diverse array of work being marketed – or mis-marketed – as 'virtual reality', it is important to engage the specific affordances of each project's particular flavour of 'VR'. The remainder of the chapter explores four different approaches:

• 360-Degree Video

This category contains projects that make use of the increasing ease of shooting 360-degree live-action video. Exemplars include work created by Chris Milk and other artists in partnership with his company VRSE, which is responsible for a series of pieces in which the viewer uses a computer, tablet, phone or VR headset to witness a 360-degree live-action scene with the viewer at the centre. The viewer's actions do not



Evolution of Verse (2015) by Chris Milk and VRSE

affect the project or the time base of the story; however, one is free to explore any part of the spherical image by means of a tablet, phone or head-mounted display.

One of the first projects made available from VRSE is titled *Evolution of Verse* (2015), directed by Milk and made in collaboration with Annapurna Pictures and Digital Domain. The three-minute experience features a pre-rendered 360-degree CG environment of a lake surrounded by mountains at dawn; soft piano music plays and birds chirp. Then, in what must be an homage to the Lumière Brothers, a train comes rushing through the water toward the viewer, and as it approaches impact, it explodes into a flock of black birds that swirl up and away into the bright blue sky; the birds in turn become colourful streamers in a glorious visual display as the sun begins to set behind the hills; the image fades to black and then, in what must be another homage, this time to *2001: A Space Odyssey* (1968), we see a plump baby, ostensibly a foetus floating unencumbered by something as mundane as a maternal body and the confines of a womb and illuminated from behind, umbilical cord translucent and floating upward. The baby turns, smiles and reaches its hand outward, seemingly pushing through the screen.

The dual gestures here toward the origins of cinema and the origins of man endeavour to bolster a new technology and at once underscore familiar repetitions of contemporary immersive media while sustaining the gender politics of contemporary mainstream cinema. Depictions of the foetal imagery, especially showing babies as if they are suspended in outer space, were well critiqued in the 1980s as feminists responded in particular to the now nostalgic notions of humans as disconnected, solitary individuals (see Petchesky 2000), the erasure of the maternal body, and the cultural work done in aligning the foetus in utero with the man in space, floating free (see Rothman 1986). The 'man in space' is now the immersed viewer, disconnected and solitary, floating free but for the bothersome tether. The previous generation of work created by artists such as Laurel and Davies and their investigations of the ways

in which we might design immersive experiences that include the body in compelling ways has been lost to history; *Evolution of Verse* both demonstrates the strangely obstinate attempt to remain within the ocular regime, and inadvertently thematises the political position this disembodied vision represents. No wonder so many of the subsequent VRSE projects would strive to assert a direct connection to other humans!

Indeed, in November 2015, VRSE partnered with *The New York Times* to explore the possibilities of immersive media in the context of the newspaper. On 5 November, the paper released a series of projects documenting the lives of three refugee children in the context of sixty million people displaced by war. Hana is a twelve-year-old Syrian girl who lives and works in Lebanon with her family in a refugee camp, not knowing when or if they will be able to go home; nine-year-old Chuol is from the South Sudan and has fled from his home with his grandmother and exists by hiding in a crocodile-filled swamp; and eleven-year-old Oleg lives in his rubble-strewn town, destroyed by war, in the Ukraine. The stories are described as 'virtual-reality films' but are, again, 360-degree videos. To facilitate the viewing of the projects and the paper's larger NYT VR initiative, the newspaper distributed Google's cardboard viewers in the 8 November 2015 issue of the paper; the Google viewers closely resemble the DIY cardboard viewer called the FOV2GO created by participants in USC's MxR Lab three years earlier and distributed for free at the 2012 IEEE Virtual Reality Conference.

The three 360-degree videos resemble traditional video shorts in form and content; cuts are enabled by dips to black, and shots are composed at a medium distance from the action to encourage the viewer to look around beyond the main subject. Indeed, the main innovation in the three projects is the ability to turn and view the surrounding environments. The height of the frame usually retains the proportions of a human adult, and the acts of turning left and right, as well as moving up and down, presume a focal point that aligns with the human head. That the projects are described as immersive suggests the breadth of experience encapsulated by the term.

These lens-based live-action 360-videos, as well as the CGI world of *Evolution of Verse*, are not responsive to the user; instead, the imagery is pre-rendered, the timeline is fixed, and the only interaction available to the viewer is to spin around to look at different parts of the captured environment. The success of projects like this is determined by the extent to which the choices available to the viewer justify the use of the 360-degree apparatus. In VRSE's spherical video document of the Millions March, directed by Spike Jonze, for example, the ability to take in the auditory and visual experience of participating in an historic protest is powerful indeed, and suggests the value of this medium for documentaries that seek to convey an experience of presence within complex, decentred events. Inverting this logic, spherical narratives that could just as easily have been conveyed through standard cinematic conventions may verge on gimmickry and ultimately detract from the intended viewer experience.

In addition to VRSE, the Oculus Story Studio, announced publicly at the Sundance Film Festival in 2015, is mining the terrain of 360-video experience. A relatively small, in-house production unit of the VR hardware manufacturer, Oculus, OSS is devoted

to a cinematic mode of production that is nearly unique among the current generation of commercial VR content creation. Where a majority of industrial production is dominated by the games industry, key players on the technical and creative teams at OSS, including Saschka Unseld, Ramiro Lopez Dau and Max Planck, are all veterans of Disney's Pixar animation studio. As such, the studio is committed to inventing a new language for immersive storytelling that nonetheless retains the characteristically nuanced graphics of a high-end animation studio while respecting the severe limitations of real-time image rendering by means of a game engine.

With their first two shorts, OSS stuck with easily recognisable genres and simple, short-form stories. Each project begins with a gentle introductory period designed to allow viewers to ease into the experience; we are invited to watch the curtains part and the lights go down in a cinema theatre, for example. With its debut short, *Lost* (2015), viewers find themselves in a darkened forest just before daybreak; a lone firefly flutters through the dark, capturing and directing the viewer's attention and allowing just enough time for the sense of presence in a lush tropical environment to give way to curiosity about the story that will unfold. As the narrative emerges, involving a robot named N8 (pronounced 'Nate'), viewers find little reason to take advantage of the 360-degree story world that surrounds them. Indeed, *Lost* treats the environment outside of a conventional narrative proscenium as an unnecessary distraction from the intended viewing experience; while viewers can choose to focus on specific figures or parts of the story as it unfolds, there is virtually never a reason to look away from the primary action or express curiosity about the surrounding environment.

A focus on narrative action and character also drives OSS's second short project, *Henry* (2015), a story about a lovable hedgehog who wants nothing more than to hug a balloon on his birthday. Here, the legacy of the team's Pixar roots is easily recognisable as Henry manages to evoke a range of emotions, from joy to sorrow. Like *Lost*, viewers have no appreciable control over either the events or the time base of the story. During the 'settling in' period, for example, time variations are limited to slightly shorter or longer durations, depending on the viewer's behaviour – an agitated viewer (as determined by head movements) will be given more time to settle into a proper viewing state, while a calm one will be rewarded with commencement of the action. Opportunities for viewers to procedurally affect the actions of the story are likewise kept to a bare minimum; at key moments, a viewer who is paying close attention to the correct part of the screen may be rewarded with eye contact and an intimate, wordless exchange with Henry at a key moment in the story.

Unlike the majority of game-oriented VR content producers, OSS is not looking to create the killer app that will drive sales of one particular delivery system over its rivals. The studio has a different mandate, namely to bring artistry and inspiration to VR as a medium of creative expression and engaging storytelling. The problem is that, in so doing, it is fighting against the very properties that make immersive experiences unique, treating the prospect of user agency and meaningful interaction as distractions rather than creative variables to embraced or shaped. In the end, OSS may well

discover a successful formula for the language of storytelling in VR, but in all likelihood, it will look a lot like the good old language of cinema.

• **Navigable Environment: Pre-Rendered CGI**

This category of VR allows the participant to move within the space of the project which is pre-rendered. This terrain has been effectively mobilised in the work of Nonny de la Peña, one of our students in the Media Arts + Practice Ph.D. programme at USC. Given her background as a journalist, de la Peña has named her body of work ‘immersive journalism’, and she is dedicated to exploring the ways in which the elements of journalistic storytelling and VR can come together, particularly with regard to empathy. Her research asks how empathy might be affected as participants engage with real-world stories.

De la Peña begins her projects with audio from real events; she then adds corresponding computer-generated images. In the seven-minute project titled *Hunger in Los Angeles* (2012), for example, an audio recording of people standing in a food bank line in downtown LA provides the framing. The 3D visual world and animated characters were rendered in Unity, prompting many to refer to the project as a game. But *Hunger in Los Angeles* is not a game; it is a navigable space in which a participant, wearing a head-mounted display equipped with markers, is able to look and listen while walking around within a defined volume. Half-way through the experience, a person in line suffers a diabetic seizure and collapses; the other people in line try to help but they are also still hungry and anxious not to lose their places in line. In the distance, the sound of a siren blares. The participant exploring the project is able to walk among the people in line, and can even hover next to the suffering victim, but, like a ghost, remains powerless to act. In this way, the immersiveness of the project creates a profound experience of care, sympathy and immediacy, but juxtaposes it with helplessness. Even though the visuals are low-resolution and the animations are non-naturalistic, the audio, and the knowledge of the audio’s veracity, prompt the sense of empathy that de la Peña seeks to achieve. Made in 2012, still the early days of the current VR wave, the project demonstrates the ability of the still unwieldy media technology to craft a significant experience.

Each of de la Peña’s subsequent projects similarly steps into a dramatic situation, placing participants in the story world to serve as a witness to the events unfolding. Each uses audio recordings as the anchor and framing device, and each allows some degree of physical movement (depending on the installation of the project). *Project Syria* (2013) places participants on a street in Aleppo in Syria, where a child sings just as a bomb explodes nearby. *One Dark Night* (2015) recreates scenes related to the last moments of Trayvon Martin’s life, and *Across the Line* (2015) brings participants into the midst of an anti-abortion protest outside a health clinic.

De la Peña is very much concerned with presence, and has been working with Mel Slater, who has spent many years conducting research on presence, immersion and sense of place in relation to virtual reality. In ‘A Note on Presence Terminology’,



Hunger in Los Angeles (2012) by Nonny de la Peña, composite image by Peggy Weil

Slater distinguishes between immersion and presence, using immersion to refer to an objective assessment of what a technology delivers: ‘The more that a system delivers displays (in all sensory modalities) and tracking that preserves fidelity in relation to their equivalent real-world sensory modalities, the more that it is “immersive”’ (2003: 1). Presence, in contrast, is for Slater ‘a human reaction to immersion’ (ibid.). The experience of presence is subjective. Slater goes on to show how the form of a project and its content relate to presence as well; we may become interested or involved in a project, but that refers to its content. Our relationship to the project with regard to presence centres on its form: ‘Presence is about form, the extent to which the unification of simulated sensory data and perceptual processing produces a coherent “place” that you are “in” and in which there may be the potential for you to act’ (2003: 2). De la Peña’s projects are unique in that they rely primarily on sound and physical movement to

produce a coherent sense of place, when the majority of VR producers focus primarily on the visual register. Hailed also for their ability to produce a sense of presence, again, despite the limited visual sensory data, de la Peña's projects suggest that the visual need not be the primary sensory modality for VR. Taken together, the success of de la Peña's work is a testament to the capacity of audio to override the visual in producing a sense of presence.

Many other artists exploring VR share de la Peña's interest in the notion of empathy. BeAnotherLab, an interdisciplinary team of artists and researchers, for example, created a project titled *The Machine To Be Another* (2015) as an experiment to explore physical identity. It uses the Oculus Rift, multiple cameras and human actors to create situations in which a participant can seem to experience another body. Once engaged in the project, the participant looks down to see different leg that are different from his or her own, and a torso, hands and arms; the goal is to spark a sense of curiosity about what it might be like to live in this body. In some ways, the project builds on the work of Jaron Lanier from the 1980s when he explored the extent to which virtual bodies could be made as strange as possible but still be accepted by participants as their own. Lanier dubbed this 'homuncular flexibility' (2006), and it is a topic currently being investigated by Stanford's Virtual Human Interaction Lab, among others.

• Navigable Environments With Temporal Flexibility

Modestly billed as a 'walk in the woods', *Way to Go* (2015) was produced by the National Film Board of Canada and directed by Vincent Morisset and debuted at the Sundance Film Festival in 2015. Offering viewers an experience that lasts anywhere from six minutes to 'maybe forever', the project models a uniquely elastic but unidirectional timeline that allows viewers a surprising degree of freedom simply by being able to control their speed. As the project starts, we are invited to follow a simple animated figure with an expressionless, cube-shaped head as it walks, runs or flies through a forest along a path next to a stream. For the first few minutes, the viewer's progress is limited to a single path signified by a white line on the ground; we can go fast, slow or stop, but we can't go back. The space of the forest is pictured in nearly monochrome, spherical, video and as we look around in any direction, the cube-head figure remains framed in our field of view. As we move further into the space, the live-action images give way to an entirely computer-generated 3D environment with an increasingly saturated colour scheme that we understand might just go on forever.

Viewable either in a browser or an Oculus Rift, standard controls allow participants to walk, run or stop, all the while being able to look around the spherical forest environment. If our gaze tilts up above the horizon, the cube-head takes to the air and soars overhead until we look back down. In addition to the animated figure, we quickly notice that we are accompanied by a black-clad live-action figure who also has a cube for a head. We realise that this must be our camera operator, but his or her presence is ingeniously integrated into the diegesis of the world by virtue of having a cube for a head and the fact that, in the course of our walk, we encounter multiple other

figures with cube-shaped heads, both animated and live-action, who accompany us for portions of our adventure.

Although lacking anything that could be regarded as a game mechanic, *Way to Go* rewards viewers for their curiosity and patience. Further, while it would be possible to simply race through the woods, running or flying as quickly as possible, viewers who pause occasionally or allow themselves to explore the environment will be rewarded by unexpected pleasures. At one point, the cube-head strays off the path that we, as viewers of a live-action video sequence, know we cannot leave, and disappears below the surface of the stream. If we stop and simply observe the running water, we see the cube head staring back at us from under the water until the stream itself bursts into a rainbow of colour that rushes downstream. Further on, the live-action video sequences and the narrow path we are constrained to follow give way to an entirely computer-generated forest environment that pulses with psychedelic colours and textures reminiscent of the video world in which we were previously immersed.

However, with *Way to Go*, it is at the intersection of live-action, animation and CG that the project achieves something unique and exemplary for the current generation of VR content. By keeping narrative elements – and viewer expectations – to a minimum, a deceptively simple environment is shown to conceal a wealth of treasures and surprises. By limning the boundaries between human and animated character, live-action and procedurally generated images, and by gently bending the ordinarily immutable limitations of live-action video, *Way to Go* hints at the potential of this medium to rethink the nature of both story and immersive experience.

While explorations of the landscape form a genre in contemporary VR – Rachel Rossin's *I Came and Went as a Ghost Hand* (2015) explores glitchy landscapes situated between the virtual and real and Daniel Steegmann Mangran's *Phantom (kingdom of all the animals and all the beasts is my name)* (2015) invites participants to explore the Mata Atlântica rainforest in Southern Brazil both come to mind – *Way to Go*'s flexible timeline hints at one direction for imagining the future of storytelling within VR. While films within traditional cinematic storytelling unspool at a fixed speed, with a good portion of cinema's artistry devoted to choreographing the temporal unfolding, what if we imagined a form of storytelling that allowed time to be stopped, rewound or fast-forwarded as part of the narrative design?⁷

So it would seem that cinema is striving to be VR as VR is striving to be cinematic and both are contending with a world that has shifted from images and the realm of representation to data and the realm of the network. The shift, however, is uncanny: it cheerfully barrels forward at breakneck pace in some realms but remains anxiously repressed in others such that we do not even yet share a language for talking about virtual cameras. A networked culture needs, wants and crafts stories of the networked self, requiring differing methods for rendering the deeply layered spatial and temporal experiences we share daily. Whereas the impetus for many VR projects in the 1990s centred on a reimagining of the role of the body, projects fifteen years later combine

the post-cinematic and the posthuman in order to conjure and contemplate distributed bodies and an awareness of the growing sentience of the world around. The camera itself registers that sentience metaphorically in so many films, and the participant within the flourishing array of VR projects becomes the camera, registering the space and time of the story world, tracing the path of a narrative with the body. While *The Revenant*, in its diminution of narrative, imagines the hero alone, the post-cinematic and posthuman, at its best, will in contrast increasingly explore the relationships among people and things. Returning to the definition of the post-cinematic from the beginning of this book – the art and practice of bringing experiences of story, information and knowledge into attunement with contemporary culture – the qualities of VR that align it with the post-cinematic and the posthuman will be those that allow it to thrive.